

Computer Vision - Exercise 3

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This exercise will be discussed the week of 18th of May.

1 Introduction

The following exercises cover local features, edge detection, scale-space blob detection, and feature descriptors.

2 Exercises

Exercise 1: Image Gradients and Edge Orientation

An image edge can be characterized as a location of rapid intensity change. Let the local image gradients at a pixel be given by

$$g_x = 3, \quad g_y = 4.$$

1. Compute the gradient magnitude.
2. Compute the gradient orientation

$$\theta = \text{atan2}(g_y, g_x).$$

3. Explain why large gradient magnitudes are useful for edge detection.
4. In the Canny edge detector, why is non-maximum suppression applied after computing the gradient magnitude?

Exercise 2: Laplacian Filter

The discrete Laplacian filter is

$$D = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Consider the image patch

$$I = \begin{pmatrix} 80 & 80 & 80 \\ 80 & 120 & 80 \\ 80 & 80 & 80 \end{pmatrix}.$$

1. Compute the Laplacian response at the center pixel.
2. Explain why the Laplacian filter responds strongly to isolated bright or dark spots.
3. In practice, computing the true Laplacian of a Gaussian-smoothed image (LoG) is computationally expensive. How is this operator typically approximated in modern computer vision to improve efficiency?

Exercise 3: Scale-Space and Blob Detection

The Laplace operator is defined as the sum of unmixed second spatial derivatives:

$$\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}.$$

The **Laplacian of Gaussian (LoG)** applies this operator to a Gaussian-smoothed image, $L(x, y; \sigma) = \Delta(G_\sigma * f)$.

1. Explain why the LoG operator is suitable for blob detection.
2. What is meant by finding extrema of the LoG in *scale space*?
3. Define the concept of *characteristic scale*.
4. Why is it important to search for extrema not only in space (x, y) but also across scales σ ?